

Exercises

1. By Exercise 35 in §4.2 there is a bundle $S^2 \rightarrow \mathbb{CP}^3 \rightarrow S^4$. Let $S^2 \rightarrow E_k \rightarrow S^4$ be the pullback of this bundle via a degree k map $S^4 \rightarrow S^4$, $k > 1$. Use the Leray-Hirsch theorem to show that $H^*(E_k; \mathbb{Z})$ is additively isomorphic to $H^*(\mathbb{CP}^3; \mathbb{Z})$ but has a different cup product structure in which the square of a generator of $H^2(E_k; \mathbb{Z})$ is k times a generator of $H^4(E_k; \mathbb{Z})$.
2. Apply the Leray-Hirsch theorem to the bundle $S^1 \rightarrow S^\infty/\mathbb{Z}_p \rightarrow \mathbb{CP}^\infty$ to compute $H^*(K(\mathbb{Z}_p, 1); \mathbb{Z}_p)$ from $H^*(\mathbb{CP}^\infty; \mathbb{Z}_p)$.
3. Use the Leray-Hirsch theorem as in Corollary 4D.3 to compute $H^*(V_n(\mathbb{C}^k); \mathbb{Z}) \approx \Lambda_{\mathbb{Z}}[x_{2k-2n+1}, x_{2k-2n+3}, \dots, x_{2k-1}]$ and similarly in the quaternionic case.
4. For the flag space $F_n(\mathbb{C}^n)$ show that $H^*(F_n(\mathbb{C}^n); \mathbb{Z}) \approx \mathbb{Z}[x_1, \dots, x_n]/(\sigma_1, \dots, \sigma_n)$ where σ_i is the i^{th} elementary symmetric polynomial.
5. Use the Gysin sequence to show that for a fiber bundle $S^k \rightarrow S^m \xrightarrow{p} S^n$ we must have $k = n - 1$ and $m = 2n - 1$. Then use the Thom isomorphism to show that the Hopf invariant of p must be ± 1 . [Hence $n = 1, 2, 4, 8$ by Adams' theorem.]
6. Show that if M is a manifold of dimension $2n$ for which there exists a fiber bundle $S^1 \rightarrow S^{2n+1} \rightarrow M$, then M is simply-connected and $H^*(M; \mathbb{Z}) \approx H^*(\mathbb{CP}^n; \mathbb{Z})$ as rings. Conversely, if M is simply-connected and $H^*(M; \mathbb{Z}) \approx H^*(\mathbb{CP}^n; \mathbb{Z})$ as rings, show there is a bundle $S^1 \rightarrow E \rightarrow M$ where $E \simeq S^{2n+1}$. [When $n > 1$ there are examples where M is not homeomorphic to $\mathbb{CP}^n$.]
7. Show that if a disk bundle $D^n \rightarrow E \rightarrow B$ has a Thom class with $\mathbb{Z}$ coefficients, then it is orientable.
8. If E is the product bundle $B \times D^n$ with B a CW complex, show that the Thom space $T(E)$ is homotopy equivalent to the n -fold reduced suspension $\Sigma^n B$, and that the Thom isomorphism specializes to the suspension isomorphism $H^i(B; R) \approx \tilde{H}^{n+i}(\Sigma^n B; R)$ given by the reduced cross product in §3.2.
9. Show that the inclusion $T^n \hookrightarrow U(n)$ of the n -torus of diagonal matrices is homotopic to the map $T^n \rightarrow U(1) \hookrightarrow U(n)$ sending an n -tuple of unit complex numbers $(z_1, \dots, z_n)$ to the 1×1 matrix $(z_1 \cdots z_n)$. Do the same for the diagonal subgroup of $Sp(n)$. [Hint: Diagonal matrices in $U(n)$ are compositions of scalar multiplication in n lines in $\mathbb{C}^n$, and $\mathbb{CP}^{n-1}$ is connected.]
10. Fill in the details of the following argument to show that every $n \times n$ matrix A with entries in $\mathbb{H}$ has an eigenvalue in $\mathbb{H}$. (The usual argument over $\mathbb{C}$ involving roots of the characteristic polynomial does not work due to the lack of a good quaternionic determinant function.) For $t \in [0, 1]$ and $\lambda \in S^3 \subset \mathbb{H}$, consider the matrix $t\lambda I + (1-t)A$. If A has no eigenvalues, this is invertible for all t . Thus the map $S^3 \rightarrow GL_n(\mathbb{H})$, $\lambda \mapsto \lambda I$, is nullhomotopic. But by the preceding problem and Exercise 10(b) in §3.C, this map represents n times a generator of $\pi_3 GL_n(\mathbb{H})$.

4.E The Brown Representability Theorem

In Theorem 4.58 in §4.3 we showed that Ω -spectra define cohomology theories, and now we will prove the converse statement that all cohomology theories on the CW category arise in this way from Ω -spectra.

Theorem 4E.1. *Every reduced cohomology theory on the category of basepointed CW complexes and basepoint-preserving maps has the form $h^n(X) = \langle X, K_n \rangle$ for some Ω -spectrum $\{K_n\}$.*

We will also see that the spaces K_n are unique up to homotopy equivalence.

This theorem gives another proof that ordinary cohomology is representable as maps into Eilenberg–MacLane spaces, since for the spaces K_n in an Ω -spectrum representing $\tilde{H}^*(-; G)$ we have $\pi_i(K_n) = \langle S^i, K_n \rangle = \tilde{H}^n(S^i; R)$, so K_n is a $K(G, n)$.

Before getting into the proof of the theorem let us observe that cofibration sequences, as constructed in §4.3, allow us to recast the definition of a reduced cohomology theory in a slightly more concise form: A reduced cohomology theory on the category $\mathcal{C}$ whose objects are CW complexes with a chosen basepoint 0-cell and whose morphisms are basepoint-preserving maps is a sequence of functors h^n , $n \in \mathbb{Z}$, from $\mathcal{C}$ to abelian groups, together with natural isomorphisms $h^n(X) \approx h^{n+1}(\Sigma X)$ for all X in $\mathcal{C}$, such that the following axioms hold for each h^n :

- (i) If $f \simeq g: X \rightarrow Y$ in the basepointed sense, then $f^* = g^*: h^n(Y) \rightarrow h^n(X)$.
- (ii) For each inclusion $A \hookrightarrow X$ in $\mathcal{C}$ the sequence $h^n(X/A) \rightarrow h^n(X) \rightarrow h^n(A)$ is exact.
- (iii) For a wedge sum $X = \bigvee_{\alpha} X_{\alpha}$ with inclusions $i_{\alpha}: X_{\alpha} \hookrightarrow X$, the product map $\prod_{\alpha} i_{\alpha}^*: h^n(X) \rightarrow \prod_{\alpha} h^n(X_{\alpha})$ is an isomorphism.

To see that these axioms suffice to define a cohomology theory, the main thing to note is that the cofibration sequence $A \rightarrow X \rightarrow X/A \rightarrow \Sigma A \rightarrow \dots$ allows us to construct the long exact sequence of a pair, just as we did in the case of the functors $h^n(X) = \langle X, K_n \rangle$. In the converse direction, if we have natural long exact sequences of pairs, then by applying these to pairs of the form (CX, X) we get natural isomorphisms $h^n(X) \approx h^{n+1}(\Sigma X)$. Note that these natural isomorphisms coming from coboundary maps of pairs (CX, X) uniquely determine the coboundary maps for all pairs (X, A) via the diagram at the right, where the maps from $h^n(A)$ are coboundary maps of pairs and the diagram commutes by naturality of these coboundary maps. The isomorphism comes from a deformation retraction of CX onto CA . It is easy to check that these processes for converting one definition of a cohomology theory into the other are inverses of each other.

$$\begin{array}{ccc} h^n(A) & \xrightarrow{\quad} & h^{n+1}(X/A) \\ \downarrow & \searrow & \uparrow \\ h^{n+1}(CA/A) & \xleftarrow{\approx} & h^{n+1}(CX/A) \end{array}$$

Most of the work in representing cohomology theories by Ω -spectra will be in realizing a single functor h^n of a cohomology theory as $\langle -, K_n \rangle$ for some space K_n .